

RYAN HACKNEY

Electrical & Computer Engineer | Full-Stack Developer | Cybersecurity

CONTACT

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SKILLS

Languages

Python, JavaScript, TypeScript,
C/C++, Java, SQL

Web/Mobile

React, React Native, Node.js,
Express, Flask, FastAPI, Next.js

Data & ML

PostgreSQL, MongoDB, Redis,
scikit-learn, XGBoost, TensorFlow

Security

Kali Linux, Metasploit, Nmap, Pen
Testing

Hardware

PCB Design, Embedded C,
Firmware, Arduino, Raspberry Pi

Tools

Git, Docker, AWS S3, Linux,
VMware

EDUCATION

THE Ohio State University

B.S. Electrical & Computer
Engineering
Graduated 2019

CERTIFICATIONS

Google Cybersecurity Professional
Certificate (2025)

Full-stack developer with 3 production applications demonstrating React Native mobile development, ML integration, and scalable backend architecture. Hands-on penetration testing training (TCM Security Academy, Google Cybersecurity Certificate). Electrical & Computer Engineer with extensive hardware experience transitioning into software development roles.

SOFTWARE PROJECTS

Intentional Movement - Social Lifestyle Mobile App

React Native • Node.js • PostgreSQL • Stripe • Socket.io • AWS S3

- Cross-platform mobile app with 50+ API endpoints, real-time chat, payment processing, JWT auth
- Deployed on Railway with PostgreSQL database, AWS S3 cloud storage, and Stripe payment integration

Stocky - AI Stock Prediction App

React • Python/Flask • PostgreSQL • scikit-learn • XGBoost

- Ensemble ML model with 54.51% accuracy using Random Forest, XGBoost, Neural Networks
- Custom feature engineering pipeline with RESTful API and automated model retraining

NFL Predictor - Sports Analytics Platform

React • Python/FastAPI • Node.js • MongoDB

- Full-stack app with ensemble ML models and 20+ engineered features

HARDWARE & ENGINEERING

FPV Racing Quadcopter

Custom PCB • Embedded C/C++ • PID Control • RF Systems

- Built custom flight controller PCB with ARM Cortex-M microcontroller, hand-soldered SMD components
- Programmed custom firmware with PID control loops, IMU sensor fusion, failsafe systems

Autonomous Mapping Robot (Capstone)

Python • Raspberry Pi • Pathfinding Algorithms • Robotics

- Designed autonomous robot for mapping hazardous environments with ultrasonic sensors and optimal escape route calculation

3D Printer Assembly & CAD Design

Marlin Firmware • Fusion 360 • Mechanical Assembly

- Assembled MicroMax 3D printer, configured Marlin firmware and Cura slicer, installed BLTouch auto-leveling
- CAD design for functional parts (quadcopter mounts, electronics enclosures, project prototypes)

BMW 335i Performance Build

ECU Tuning • Diagnostics • Performance Optimization

- Full bolt-on modifications (400+ whp), ECU tuning with MHD, transmission tuning with XHP
- BMW ISTA diagnostics, datalogging, AFR/boost/timing analysis for performance optimization

CYBERSECURITY

TCM Security Academy - Practical Ethical Hacking

- Hands-on labs: exploited Kioptrix Level 1 and Blue (EternalBlue MS17-010) vulnerable machines
- Network enumeration (Nmap), exploitation (Metasploit), privilege escalation, post-exploitation

EXPERIENCE

Customer Service Representative - Working Solutions

Remote | Starting November 2025